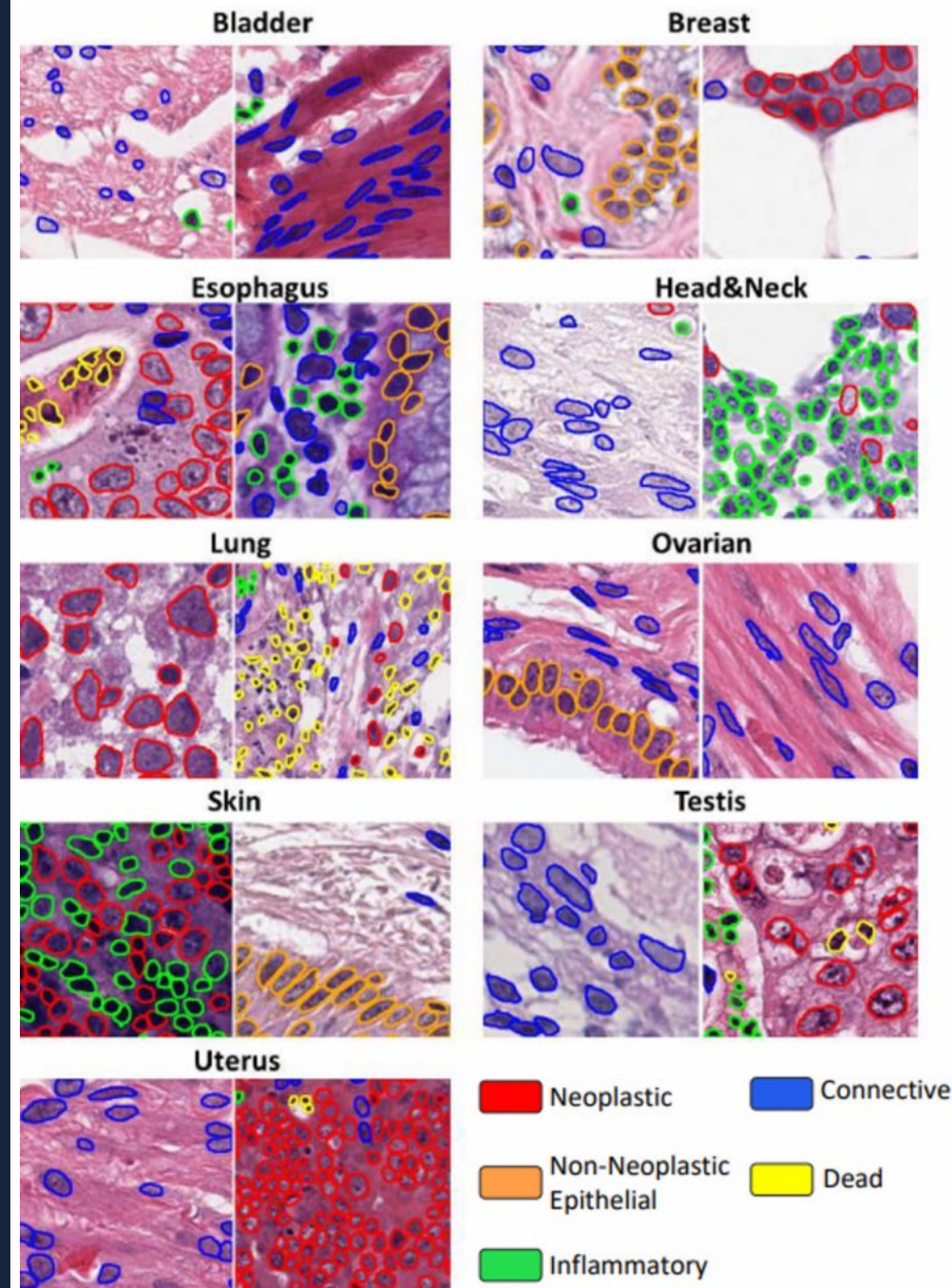


Nuclei instance segmentation across diverse tissues

One optimization route:
resolution-aware instance decoding

Results through October 5, 2026

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Can more working pixels reduce missed Dead nuclei?

Why test resolution?

Dead nuclei are small:
median area 107 native px^2
versus 368 px^2 for other classes.

One training change

Same CellViT-UNI recipe.
Working image: $256 \times 256 \rightarrow 512 \times 512$.
Evaluate instances on the original 256 grid.

All result tables: 3-split means, seed 19, no test-time augmentation (TTA).
Shared evaluator; decode parameters selected on validation only.

Upsampling changes the working grid; it does not add new acquisition detail.

Test the resolution change, then examine both small-nucleus detection and instance quality

512 helps Dead PQ, but lowers overall instance quality

Working resolution	mPQ	bPQ	Dead PQ
256 baseline	0.4995	0.6653	0.1760
512, original decoder	0.4791	0.6413	0.1816
Difference (512 – 256)	−0.0204	−0.0240	+0.0056

mPQ

Overall quality with types

bPQ

Instance quality, ignoring types

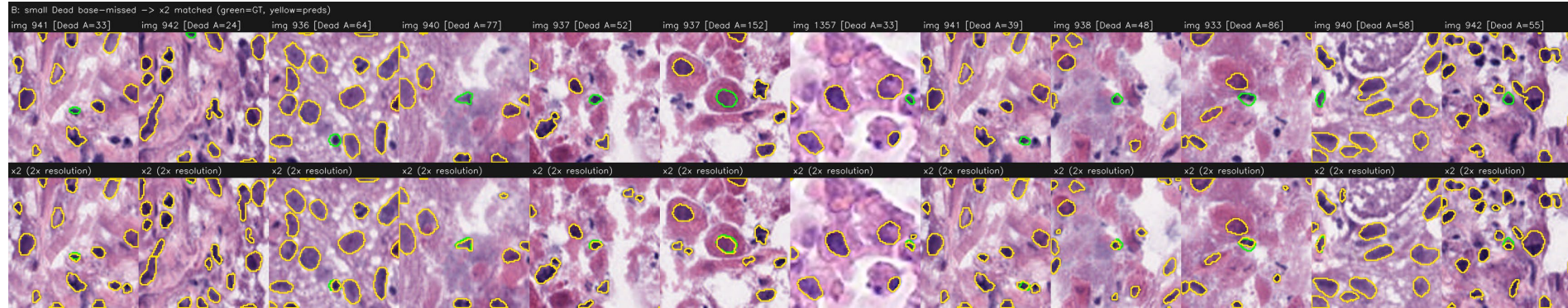
Dead PQ

Instance + type quality for Dead

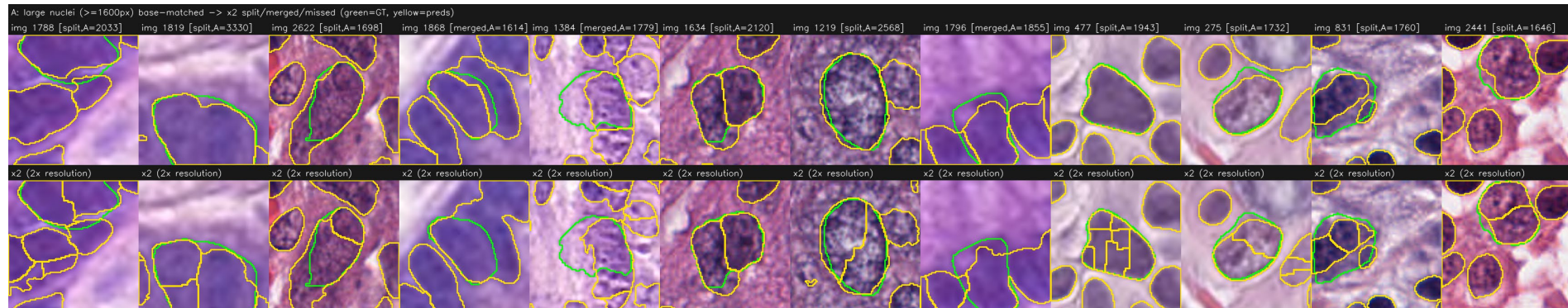
The Dead-specific score rises slightly while overall quality falls; inspect the errors

Small nuclei appear; some large nuclei break into fragments

Small-nucleus gain — each pair: 256 above, 512 below



Large-nucleus damage — multiple predictions inside one GT outline



Better small-object detection can coexist with worse separation of larger objects

Scale the decoder together with the working resolution

Decoder setting	256 baseline	512, unchanged	512 + du2
Minimum object / marker area	10 px ²	10 px ² = 2.5 native px ²	40 px ² = 10 native px ²
Sobel aperture	21 px	21 px ≈ 10.5 native px	41 px ≈ 20.5 native px

At 2× resolution: multiply area thresholds by 4 and length scales by about 2.
Re-decode the existing 512 predictions; no further training.

Keep the effective decoder scale comparable when testing a new working resolution

du2 recovers 50% of the mPQ loss and 71% of the bPQ loss

Configuration	mPQ	bPQ	Dead PQ
256 baseline	0.4995	0.6653	0.1760
512, original decoder	0.4791	0.6413	0.1816
512 + du2	0.4894	0.6583	0.1831

50%71%+0.0071

mPQ loss recoveredbPQ loss recoveredDead PQ vs baseline

Most of the decoder-related loss is recovered, but a residual aggregate penalty remains

Dead miss rate falls by 7 percentage points; PQ still has a cost

Endpoint	256 baseline	512 + du2	Difference
Interior Dead miss	37.2%	30.2%	−7.0 pp
mPQ	0.4995	0.4894	−0.0101
bPQ	0.6653	0.6583	−0.0070
Dead PQ	0.1760	0.1831	+0.0071

Interior miss: non-border Dead GT without a matched prediction. pp = percentage points.

The detection benefit is clearer than the small, seed-sensitive increase in Dead PQ

Working resolution and decoder scale must be considered together

- 1 Detection benefit** Corrected 512 reduces interior Dead miss:
37.2% → 30.2%.
- 2 Method finding** Scale pixel-based decoder settings with resolution.
du2 recovers most of the initial aggregate loss.
- 3 Remaining work** Reduce residual large-nucleus fragmentation
while keeping the decoder scale controlled.

A measured detection improvement with a remaining cost in overall instance quality